The Zero Click Horizon? Cognitive Precursors to Behavioral Changes in AI-Assisted Web Search

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Ethics statement

This study involving humans were approved by the institutions Research Ethics Board (Projet #: 2024-5926). The studies were conducted in accordance with the local legislation and institutional requirements. The participants provided their written informed consent to participate in this study.

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ABSTRACT

The integration of large language models (LLMs) into search engines, most prominently through AI-generated summaries displayed at the top of results pages, represents a significant transformation of the search ecosystem. While emerging evidence suggests these features reduce click-through rates to external websites, few studies have measured this experimentally in a controlled laboratory study, and the cognitive mechanisms underlying this behavioral shift remain poorly understood. This within-subjects experiment (N = 26) investigated the effects of AI-generated summaries on cognitive load (CL), visual attention, and click-through rate using a prototype Google Search interface and eye-tracking. Participants completed informational and product-specific search tasks under both AI-assisted and non-AI-assisted conditions, with eye-tracking data collected, along with the number of clicks on external websites and self-reported satisfaction and purchase intention. Despite no statistically significant differences in self-reported perceptions, AI-assisted search showed trend-level lower click-through rates and were rated as positive. Greater focal attention was found in the AI-assisted searches, indicating deeper engagement. Pupillometry analysis revealed a crossover pattern for the AI conditions, with reduced cognitive load early in tasks followed by increased load in later stages, relative to non-AI conditions. This suggests that AI summaries function as cognitive scaffolds that shift effort from initial orientation toward eventual verification. These findings suggest that physiological and attentional measures capture early-stage effects of AI integration that self-report metrics alone cannot detect and point toward a mechanism through which AI summaries may progressively, but negatively, impact external web traffic.

KEYWORDS: Search Engines, User Experience, User Interaction, Generative AI, Artificial Intelligence, Eye Tracking

1. INTRODUCTION

With billions of queries processed daily, search engines (SEs) serve as critical gateways to knowledge, news, goods, and services on the Internet. Although their primary function is to facilitate access to relevant information in response to user queries, SEs are not passive conduits. They employ increasingly sophisticated ranking algorithms that curate content based on reliability, keywords, and popularity, shaping which information users encounter (Beel & Gipp, 2009; Hingoro & Nawaz, 2021; Seymour et al., 2011). Crucially, the majority of users confine their attention to the first page of results, often to the first several links alone. This attentional bottleneck has given rise to a dedicated field, Search Engine Optimization (SEO), in which businesses and digital platforms invest considerable effort to secure prominent placement in search results (Almukhtar et al., 2021; Bhandari & Bansal, 2018; Matošević et al., 2021; Sharma et al., 2019).

This placement is not merely desirable; for many businesses, it is existential. The resulting dynamic is one of deep mutual dependence: SEs depend on the continuous creation of online content, while content creators rely heavily on SEs to reach their audiences (Graham et al., 2017; Van Couvering, 2011; Vincent et al., 2019). The economic scale of this interdependence is substantial. In the second quarter of 2024, Google's revenue exceeded 84.2 billion U.S. dollars, with the company achieving a record annual revenue of 305.63 billion dollars in 2023, the majority driven by advertising on Google sites and its network (Statistica, 2024). Because advertisers primarily pay on a cost-per-click model, the click-through rate (CTR) in search results directly governs revenue. Any alteration in SE functionality that shifts CTR therefore carries significant economic consequences for both the search engines themselves and the ecosystem of businesses that depend on them.

1.1 AI Transforming Search Engines

Over the past decade, SEs have evolved beyond the display of ranked links by incorporating features such as sponsored links, highlighted snippets, and frequently asked questions (Bink et al., 2022; Strzelecki & Rutecka, 2020). However, the most consequential transformation is now being driven by artificial intelligence (AI), particularly by Large Language Models (LLMs). These models learn from vast text corpora, often comprising billions of data points, and have demonstrated strong capabilities in natural language understanding, text generation, and summarization (Patil & Gudivada, 2024; Raiaan et al., 2024). LLMs are being applied across diverse sectors (Biswas, 2023; Chodak & Błażyczek, 2023), including SEO optimization (Chodak & Błażyczek, 2023; Yuniarthe, 2017), and some studies suggest they could serve as a primary method for information retrieval, potentially displacing SEs altogether (Karunaratne & Adesina, 2023; Lindemann, 2023; Shen et al., 2024; Strzelecki, 2020).

Rather than replacing SEs, however, the predominant approach has been to integrate LLMs into existing search infrastructure. AI models have long contributed to web page ranking algorithms (Boyan et al., 1996; Yoganarasimhan, 2020), but recent efforts go further, embedding LLMs directly into the search interface and recommendation page (Maoro et al., 2023). Microsoft has incorporated its AI model, Copilot, into Bing (Blogs, 2023), and Google, the most widely used SE by a significant margin, is integrating Gemini into Google Search as part of its AI-powered Search Generative Experience (SGE) (Google, 2023a, 2023b, 2024a). The most common form of this integration is the generation of AI summaries displayed prominently at the top of search results (see Figure 1), a feature Google terms 'AI Overviews' (Google, 2024b). The stated goal is to enhance user experience and supplement web traffic by providing an alternative pathway to information. Although these AI-generated results do not yet contain advertisements, search engines are testing user engagement with the feature in anticipation of future advertising integration and revenue creation.

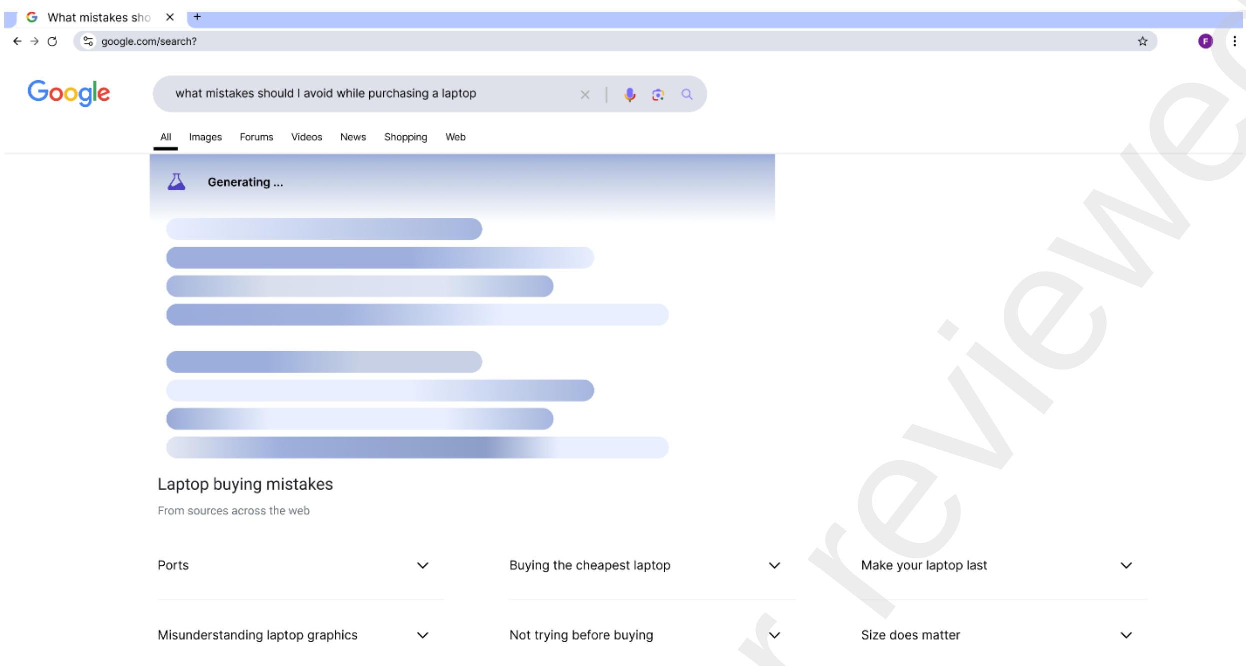

Figure 1. Example of a summary being generated following a query in Google's AI-powered Search Generative Experience known as ‘AI Overviews’.

1.2 Disruption of the Search Ecosystem

These developments have raised substantial concerns among content creators, businesses, and advertisers, who fear that AI-generated summaries will redirect user attention away from linked websites, thereby disrupting the economic interdependence described above (Bag et al., 2022; Gkikas & Theodoridis, 2022; Theodoridis & Gkikas, 2019). If AI summaries satisfy users' information needs directly, the result would be fewer visits to external websites and fewer opportunities for monetization through advertising and sponsored content. Emerging evidence has begun to validate these concerns. Prior research on AI-assisted search engines, such as Bing Chat, found that they cite and link to fewer external websites compared to standard SEs (Kelly et al., 2023). More recently, Dewanto and Herari (2025) documented a rise in zero-click searches, in which users satisfy their queries without clicking any external links. Law and Guan (2025) report that the presence of an AI Overview correlates with a 34.5% reduction in CTR for the top-ranking page. Furthermore, Kaiser et al. (2025) show that users are more likely to engage with AI-generated search results than with traditional listings. In contrast, Yin et al. (2025) suggest that AI-personalized recommendations can boost click intention in commercial settings. However, this effect depends heavily on ‘immersive experience’, which may paradoxically keep users engaged with the AI interface itself rather than directing them to external links.

1.3 Moving from Behavioral Observation to Cognitive Mechanisms

While the potential of AI-generated summaries to reduce traffic is becoming empirically established, the specific cognitive mechanisms and user behaviors driving this change remain poorly understood. It is not yet clear, for example, whether summaries influence clicking behavior by satisfying information needs preemptively, by altering the perceived relevance of linked results, or by some other pathway. Nor is it known whether users find these summaries genuinely useful, whether they are simply consumed passively, or whether they are skipped entirely, as advertising links often are (Jerath et al., 2011). Furthermore, the broader effects on user perceptions, including satisfaction and perceived usefulness remain largely unexplored.

Addressing this gap requires moving beyond aggregate behavioral metrics like CTR to examine the underlying cognitive processes that may precede observable changes in behavior. Doing so, however, requires a framework capable of addressing three distinct questions: how do users process the additional information introduced by AI summaries, where do they direct their attention within the modified interface, and why do immediate behavioral metrics fail to register effects that are already in process at a cognitive level. We draw upon three theoretical perspectives, from cognitive psychology and behavioral science, to investigate these questions and, when taken together, provide the analytic framework for the present study.

1.4 Theoretical Framework: Cognitive Load, Visual Attention, and Habit

Firstly, Cognitive Load Theory (CLT) posits that human information processing is constrained by the limited capacity of working memory, and that the mental effort required to process information is a key determinant of decision quality and engagement (Sweller, 2011). In the context of search, AI-generated summaries may affect cognitive load in opposing directions. Concise, well-structured summaries could reduce load by providing direct answers and lowering information density, potentially improving user experience and engagement. Nevertheless, summaries that conflict with user expectations, introduce unfamiliar framings, or add competing information to an already complex results page could increase cognitive load, impair decision-making and reduce engagement (Paas & van Merriënboer, 2020; Skulmowski & Xu, 2022). Studies have shown that various aspects of online content, including content type, complexity, presentation, and information volume, influence cognitive states, which in turn affect user engagement and behavior (Lewandowski & Kammerer, 2021; Skulmowski & Xu, 2022; Wang et al., 2014). High cognitive load, in particular, can impair decision-making, reduce engagement, and increase user frustration. Consequently, cognitive load is a plausible mediator between the introduction of AI summaries and any downstream changes in clicking behavior.

Secondly, visual attention research, informed by eye-tracking methodology, offers a complementary lens. Users typically concentrate their gaze on the top and left portions of search engine results pages (Lewandowski & Kammerer, 2021), precisely where AI-generated summaries are positioned. This placement may effectively capture attention and enhance engagement by delivering immediate answers in high-visibility areas. However, the same positioning introduces the risk of 'banner blindness,' a well-documented phenomenon in which users learn to ignore content placed in locations typically associated with advertisements (Benway, 1998; Jerath et al., 2011). Eye-tracking data can therefore reveal not only whether summaries attract attention, but also whether and how they are read, providing insights that aggregate click data alone cannot offer(Strzelecki, 2020).

Finally, Habit Theory suggests that search engine interactions are often governed by automaticity and repetition, where past behavior is the strongest predictor of future action, largely independent of interface changes (Ouellette & Wood, 1998; Verplanken, 2006). Introducing AI summaries into an established search workflow may therefore not produce immediate changes in clicking behavior, because habitual patterns carry considerable behavioral inertia. This observation is critical: it implies that the absence of immediate behavioral change should not be interpreted as the absence of any effect. Research has established a strong temporal relationship between visual fixation and cognitive processing (Just & Carpenter, 1980; Oberauer, 2019), indicating that shifts in attention allocation can serve as early indicators of changing engagement (Matthews et al., 2010; Wang et al., 2021). Moreover, CLT implies that if AI summaries successfully reduce the cognitive effort required to satisfy an information need, users will eventually gravitate toward this lower-effort pathway.

When considered together, these three perspectives lead to a specific prediction: if AI-generated summaries alter the cognitive landscape of search, evidence of this change should be observable in measures of cognitive load and visual attention, even if there are no significant differences in behavioral metrics such as click-through rate. In other words, cognitive and attentional shifts may be early indicators of longer-term behavioural change, detectable through physiological measures before they appear in the overall statistics on which current research relies.

1.5 Present Study

Accordingly, this study investigated the impact of AI-generated summaries in search engines on user interaction behavior and perceptions, compared to a traditional search interface. A Google Web Search prototype was developed featuring both a standard and an AI-assisted search experience. In the AI-enhanced version, AI-generated summaries were prominently displayed at the top of the first results page. Participants engaged with both interfaces, and their click rates, visual attention (measured via eye tracking), cognitive load, and self-reported perceptions were measured. By comparing these metrics across the two conditions, this research aims to advance understanding of how AI integration affects user interaction in search contexts. More broadly, it seeks to contribute to the growing literature on the role of AI in reshaping the digital marketplace and redefining the dynamics of online information access. The following sections present the methodology (Section 2), results (Section 3), and a discussion of their implications (Section 4).

2. METHODOLOGY

A within-subjects experiment was conducted to evaluate the effects of AI-generated summaries on user engagement, cognitive processing, and search behavior. Participants completed four search tasks using a prototype Google Search interface, two with AI-generated summaries (AI-assisted) and two without (non-AI-assisted). Tasks spanned both informational and product-specific search contexts. User engagement was measured through the number of link clicks and self-reported perception of satisfaction, whereas cognitive processing was assessed using eye-tracking.

2.1 Participants

Twenty-six participants were recruited from July to August 2024. Inclusion criteria required participants to be adults, fluent in English, able to read on a computer without eyeglasses, and free of skin allergies, cardiac stimulators, epilepsy, or facial paralysis. Participants ranged in age from 19 to 48 years ($M = 27.2$, $SD = 6.85$, 17 Female). Each participant was randomly assigned to one of four experimental conditions (A, B, C, or D; $n = 6, 6, 7$, and 7, respectively), which determined the order in which AI-assisted and non-AI-assisted tasks were performed. All participants completed all four tasks.

2.2 Google Search interface

A prototype Google Search interface was developed in Figma (Figma, 2024) to ensure visual fidelity while maintaining experimental control. Using a static prototype rather than live Google

Search eliminated confounding variation from personalized advertisements, fluctuating search rankings, and internet latency. The prototype consisted of nine web pages. The first replicated a standard Google Search homepage. For each of four search queries, two results pages were created: one displaying a standard results layout (non-AI-assisted; see Figure 2a) and one featuring an AI-generated summary displayed prominently at the top of the results (AI-assisted; see Figure 2b).

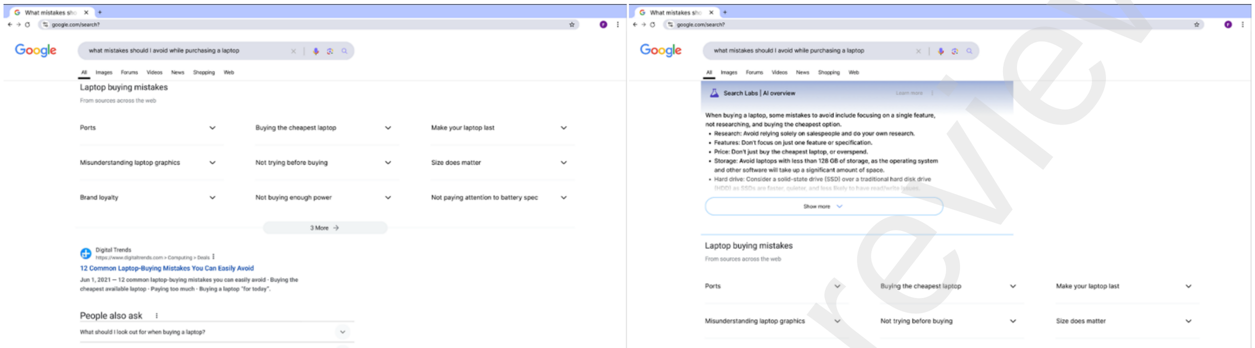

Figure 2. *left*-standard (non-AI-assisted) Google Search task for informational purposes; *right*-Google Search (AI-assisted) task for informational purposes.

The four queries were evenly divided between informational searches (e.g., "what are the important features to consider when purchasing a gaming laptop") and product-specific searches (e.g., "I want to buy a MacBook Air M1"). The two types of searches are summarized in Table 1. This design yielded four distinct task conditions: AI-assisted informational, AI-assisted product, non-AI-assisted informational, and non-AI-assisted product.

Table 1. Seach tasks

Task type	Task objective	Examples
Informative	Identify important features/criteria to consider when purchasing an item	"Use Google to find what are the important features to consider when purchasing a laptop for gaming" "Use Google to find what mistakes you should avoid doing while purchasing a laptop for gaming"
Transactional	Find a Google result that leads to a site enabling purchase of items	"Use Google to buy a Macbook Air M1" "Use Google to buy a Dell XPS 15"

2.3 Experimental Design and Procedure

Each participant completed all four tasks, but the assignment of AI assistance to specific tasks was counterbalanced across four conditions (Table 2). This ensured that every task appeared with equal frequency in AI-assisted and non-AI-assisted versions and that order effects were controlled. For example, participants in Condition A received AI assistance on Tasks 1 and 2, while participants in Condition B received AI assistance on Tasks 1 and 4.

Table 2. Experimental conditions.

Condition	Task 1	Task 2	Task 3	Task 4
A	AI	AI	NA	NA
B	AI	NA	NA	AI
C	NA	AI	AI	NA
D	NA	NA	AI	AI

Note: AI = AI-assisted searches, NA = non-AI-assisted searches

Participants provided informed consent prior to the study and were informed that their session would last approximately one hour, that they could withdraw at any time without penalty, and that they would receive $20 compensation upon completion. The experiment was conducted in a controlled laboratory environment. Each task lasted approximately five minutes, during which participants were instructed to use the prototype to find information relevant to the given query. After completing each task, participants completed a brief survey assessing perceived usefulness, and satisfaction with the search experience.

2.4 Measures

Visual attention and cognitive load were derived from gaze-tracking and pupillometric data recorded with the Tobii Nano Pro eye-tracker (Tobii AB, 2024). Specifically, visual attention was measured using the K-coefficient, a frequently used metric which determines the proportion of fixations on relevant information during saccadic movements (Krejtz et al., 2016). For each observation *i*, the coefficient is defined by equation (1):

$$(1)\ K_i = \frac{d_i - \mu_d}{\sigma_d} - \frac{a_{i+1} - \mu_a}{\sigma_a},$$

while the participant-level coefficient is defined by equation (2):

$$(2)\ K = \frac{1}{n}\sum_{i=1}^{n} K_i$$

μ and σ denote the sample means and standard deviations of fixation durations (d) and saccade amplitudes (a), respectively. Positive values indicate focal attention (long fixations followed by short saccades), while negative values indicate ambient attention (short fixations followed by long saccades). The participant-level coefficient is the mean of all K_i values across fixations.

Cognitive load was assessed via pupillometry. Where larger pupil dilations are associated with higher cognitive load (Debue & Van De Leemput, 2014; Kosch et al., 2023). Pupil size was baseline-corrected by subtracting the mean pupil diameter during a pre-task baseline period defined in equation (3):

$$(3)\ P_{\text{adjusted}} = P_{\text{raw}} - \grave{P}_{\text{baseline}}$$

Where P_{raw}= the original observed value, $\grave{P}_{\text{baseline}}$ is the mean of the variable measured during a designated baseline period, P_{adjusted}= the corrected value after removing baseline bias. In addition to pupil size, pupil range (max − min diameter, mm), blink rate (blinks per second) and saccade rate (number of saccades per second) were analyzed.

The click-through rate was measures as the *number of* clicks on search results during each task (Joachims et al., 2007). Self-reported measures were collected using 7-point Likert scales (Davis, 1989). Overall satisfaction was measured on a single scale from very unsatisfied to very satisfied. Purchase intention was measured as confidence in making a purchase decision, from very unconfident to very confident. Additionally, for AI-assisted tasks only, the perceived usefulness of the AI overview was assessed.

2.5 Data processing and analysis

Product-search and informational-search tasks were combined into AI and non-AI groups for all analyses. Eye-tracking data were preprocessed following standard steps (Carter & Luke, 2020) Pupil diameter values outside the physiological range of 0.5 mm to 7.0 mm were excluded. Missing or invalid data were removed. Eye-tracking metrics were analyzed across three temporal windows: the full task, the search phase (on the Google Search results page), and the non-search phase (on external websites). Repeated measures ANOVA were performed to analyze the effects of AI assistance on eye-tracking measures, with post-hoc comparisons. Wilcoxon signed-rank tests were used to compare self-reported measures and click counts between conditions, given the ordinal nature of the Likert data and the repeated-measures design. Spearman's rank correlation assessed the relationships between cognitive load, visual attention, and behavioral outcomes (satisfaction, purchase intention, and number of clicks). For all metrics outliers were identified and removed per metric using the interquartile range rule (1.5 × IQR) (Aguinis et al., 2013; Cousineau & Chartier, 2010; Yang et al., 2019). Results were interpreted as significant with threshold of $\alpha = 0.05$ and as a trend with $\alpha = 0.10$. Temporal analysis of pupillometry data was conducted using Generalized Additive Models (GAMs), Functional Data Analysis (FDA) and Functional Principal Component Analysis (FPCA). All temporal analyses were conducted on a normalized time base (0–100% of trial duration) to ensure comparability. First, GAMs were estimated to pupil trajectories for each condition, with smooth terms for time, and time × condition interaction to detect nonlinear shape differences, and participant- and task-level random effects. Second, FDA was applied to Savitzky–Golay-smoothed participant-level curves, allowing for pointwise t-tests and global permutation-based inference using both the maximum absolute t-statistic and the L^2 norm of the difference curve. Finally, FPCA characterized dominant modes of temporal variability across pupil trajectories.

3. RESULTS

3.1 Eye-Tracking: Aggregate Measures

Repeated measures ANOVA and pairwise results for eye-tracking data are shown in Table 3. There was a statistically significant difference in the K-coefficient across tasks, with higher values in the AI-assisted condition (M = 0.0919, SD = 0.166) than in the non-AI-assisted condition (M = -0.0208, SD = 0.182; $F = 12.68$, $p < .001$), indicating that participants adopted a more focal attentional strategy when AI summaries were present. Additionally, there was a significant main effect of condition on saccade rate, $F(1, 21) = 11.39$, $p = .003$, whereby participants in the AI condition exhibited a significantly lower saccade rate (M = 1.70, SD = 0.63) compared to the Non-AI condition (M = 1.99, SD = 0.63), $t(22) = -2.64$, $p = .015$, $d = -0.55$). Together, the increased focal attention and reduced saccade rate suggest that participants actively engaged with the AI-generated summaries rather than scanning past them.

No statistically significant difference in mean adjusted pupil size was observed across the entire task between the AI-assisted (M = −0.221, SD = 0.199) and non-AI-assisted conditions (M = −0.215, SD = 0.182; $F = 0.09$, $p = .78$), indicating no global difference in cognitive load. However, there was a significant condition × segment interaction for pupil size, $F = 5.01$, $p = .011$, suggesting that the effect of AI assistance on cognitive load differed across task phases. During the search phase, the AI condition showed marginally smaller pupil sizes (M = −0.26, SD = 0.17) than the non-AI condition (M = −0.22, SD = 0.18; $t(22) = -1.85$, $p = .078$, $d = -0.39$), suggesting a trend toward reduced cognitive load during direct interaction with the AI-enhanced results page. No significant differences were found during the non-search phase ($p = 0.47$).

Table 3. Eye-tracking data analysis across the full task, search phase and non-search phase

Phase	Metric	AI Condition		Non-AI Condition		Test Statistics		
		M	SD	M	SD	*t*	*p*	*d*
Full Task	Adjusted Pupil Size (mm)	-0.23	0.19	-0.22	0.17	-0.2	0.842	-0.05
	Pupil range	1.67	0.41	1.57	0.51	0.87	0.389	0.2
	Pupil std	0.19	0.05	0.18	0.04	0.69	0.494	0.16
	Saccade Rate (Hz)	1.66	0.68	1.9	0.75	-1.48	0.144	-0.34
	Blink Rate (Hz)	12.08	5.69	12.72	5.79	-0.48	0.633	-0.11
Search	Adjusted Pupil Size (mm)	-0.28	0.19	-0.22	0.18	-1.85	$0.078^{\dagger}$	-0.39
	Pupil range	1.54	0.46	1.35	0.45	1.92	$0.059^{\dagger}$	0.43
	Pupil std	0.18	0.06	0.17	0.04	1.1	0.275	0.24
	Saccade Rate (Hz)	1.65	0.69	2.00	0.83	-2.06	$.043^{*}$	-0.46
	Blink Rate (Hz)	11.92	6.23	12.69	5.49	-0.59	0.557	-0.13
Non-search	Adjusted Pupil Size (mm)	-0.22	0.2	-0.24	0.21	0.35	0.729	0.08
	Pupil range	1.5	0.46	1.49	0.55	0.05	0.961	0.01
	Pupil Std (mm)	0.19	0.06	0.19	0.05	0.45	0.657	0.1
	Saccade Rate (Hz)	1.78	0.73	1.96	0.81	-1.02	0.312	-0.23
	Blink Rate (Hz)	11.59	5.96	12.03	5.48	-0.35	0.727	-0.08

Note: $^{*}p < 0.05$, $^{\dagger}p < 0.1$.

3.2 Eye-tracking: Temporal Pupillometry

To examine the temporal dynamics obscured by aggregate measures, time-resolved analyses were conducted on adjusted pupil size (Figure 3a). GAMs revealed a significant effect of time on pupil size ($p < .001$) and a significant time × condition interaction ($p < .001$), indicating that AI and non-AI trials followed different temporal trajectories. The main effect of condition was not significant ($p = .23$), confirming no constant offset between conditions. Predicted trajectories (Figure 3b) showed that the AI condition exhibited smaller pupil sizes during the early portion of the trial (0–30%) and larger pupil sizes in the later portion (70–100%), highlighting a non-uniform crossover pattern. FDA *t*-tests (Figure 3c) identified 54 out of 400 timepoints where the AI–Non-AI difference was significant at α = .05, with negative effects early and positive effects late in the trial. Global permutation tests, however, did not confirm a statistically significant functional difference (max|*T*|: p = .34; L^2 norm: p = .098), indicating that the localized differences did not exceed global correction thresholds. FPCA (Figure 3d) indicated that the first three components captured 59.6% of total variance (PC1 = 34.0%, PC2 = 14.8%, PC3 = 10.9%), with peaks in component at approximately 20%, 60%, and 87% of the trial, consistent with the temporal regions where GAM and FDA analyses indicated condition-specific discrepancies.

These results indicate a temporal modulation of pupil responses rather than a uniform difference. Whereby, the early reduction and late increase in pupil dilation during AI-assisted trials suggest that AI summaries may initially reduce cognitive processing demands but subsequently require additional monitoring or integrative processing as the task progresses.

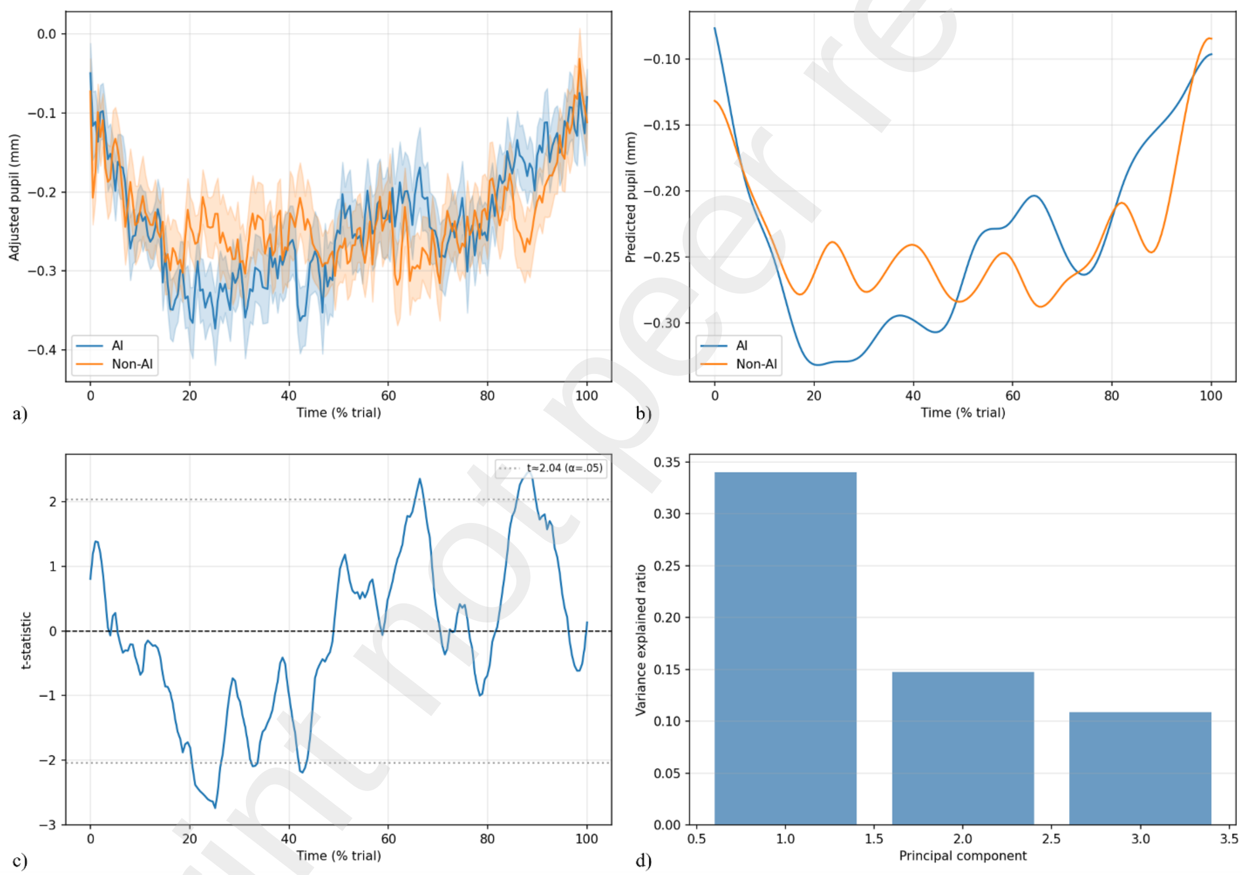

Figure 3. Time-resolved pupil dynamics across AI and Non-AI task conditions. (*a*) Average adjusted pupil size over time (±SEM), plotted as a function of trial-normalized time for non-AI assisted (orange) and AI-assisted (blue) conditions. (*b*) GAM-derived pupil trajectories illustrating nonlinear time courses for each condition; the significant time × condition smooth indicates that AI and Non-AI curves differ in shape rather than in a constant offset. (*c*) FDA time-point–wise t-statistics for the AI–Non-AI difference, demonstrating early negative and late positive deviations; dotted horizontal lines depict approximate values for *α* = .05. (*d*) FPCA variance decomposition for adjusted pupil curves, with PC1, PC2, and PC3 corresponding to early, mid, and late trial temporal patterns, respectively.

3.3 Behavioral and Self-Reported Measures

Behavioral and self-reported results are shown in Figure 4. Wilcoxon signed-rank tests reported a trend reduction in the number of clicks between from non-AI-assisted (M = 10.09, SD = 4.61) to the AI-assisted condition (M = 8.52, SD = 3.93, W = 73.0, z = -1.74, p = 0.082, r = 0.26). No significant difference in satisfaction (W = 79.0, z = -0.65, p = 0.514, r = 0.09) and intention (W = 20.5, z = -0.73, p = 0.464, r = 0.11). Perceived usefulness of AI summaries were overall positive (M = 5.53, SD = 1.29).

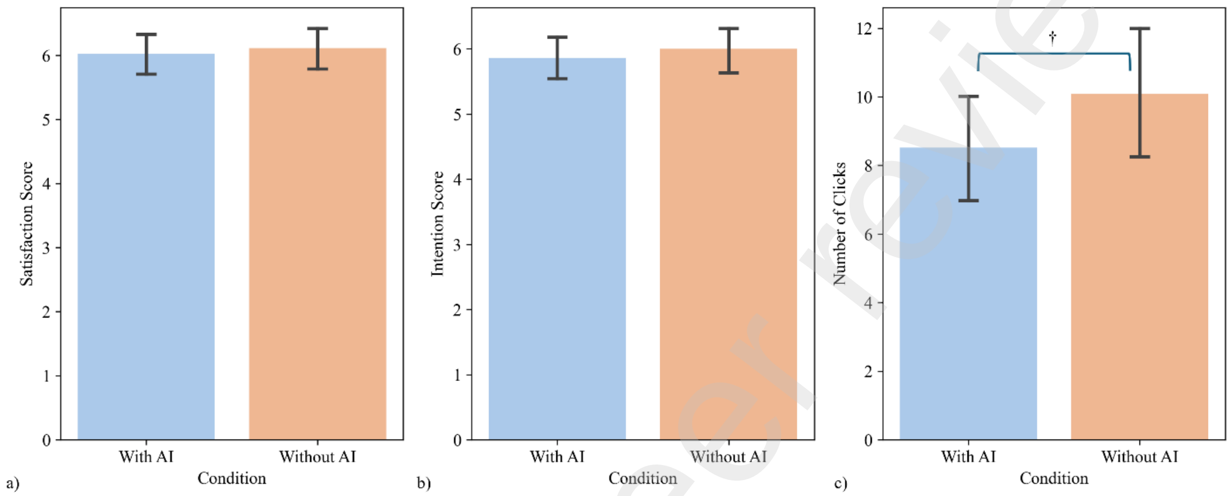

Figure 4. Average (bars) and standard deviation (errorbars) results for satisfaction (a), intention (b) and number of clicks (c) according to non-AI assisted (orange) and AI-assisted (blue) conditions, where † $p < 0.10$ from the Wilcoxon signed-rank tests.

Spearman's rank correlations were computed to determine if individual differences in cognitive processing were associated with behavioral outcomes. There was a significant negative correlation between cognitive load and number of clicks ($\rho = -0.25$, $p = .009$).Visual attention (K-coefficient) did not significantly correlate with satisfaction ($\rho = -0.11$, $p = .292$), purchase intention ($\rho = -0.19$, $p = .171$), or number of clicks ($\rho = -0.13$, $p = .19$). Similarly, cognitive load did not correlate significantly with satisfaction ($\rho = 0.17$, $p = .09$) and purchase intention ($\rho = 0.08$, $p = .57$). Self-reported data also did not correlate with number of clicks, with no statistically significant correlation between the average number of clicks and perceived usefulness, (ρ = -.08, p =.702), satisfaction, (ρ = .08, p = .692), or intention (ρ = -.01, p = .944).

4. Discussion

The integration of LLMs into search engines represents a potentially fundamental shift in information retrieval, and the present study sought to characterize its early cognitive and behavioral signatures. Specifically, it could be that AI-generated summaries altered the cognitive processes underlying visual search using evidence in measures of cognitive load and visual attention even in the absence of significant changes in behavioral metrics. There was a trend-level reduction in click-through rate when searches included an AI-generative summary, and physiological results point towards longitudinal impacts of the measured impacts on cognitive load and visual attention which could cast some light on the mechanisms underlying the broader traffic declines due to AI-generated summaries reported in recent literature (Dewanto & Herari, 2025; Kaiser et al., 2025; Law & Guan, 2025).

4.1 Main findings

The reduction in clicks in the AI condition is directionally consistent with the broader trends identified in observational studies reporting reduced CTR in the presence of AI overviews (Law & Guan, 2025). However, the effect was at a trend-level with a medium effect size, which suggests that the erosion of habitual clicking may be a gradual process. This is consistent with the predictions of Habit Theory (Ouellette & Wood, 1998; Verplanken, 2006). Search engine interactions are deeply habitual, and the behavioral inertia of established patterns may have buffered clicking behavior against the introduction of a novel interface element within a single experimental session. However, the similar of self-reported perceptions shows that users had not yet consciously registered that the AI summaries produced a qualitatively different search experience, even as their behavior and cognitive processing of that experience have measurably changed. However, participants overall rated the AI summaries as useful, even though they reported similar satisfaction and intention to purchase between conditions.

Interestingly, eye-tracking data revealed significant shifts in how users processed the search interface. The K-coefficient was significantly higher in the AI condition, indicating a transition from the ambient scanning pattern typical of standard search to a more focal attentional strategy. Concurrently, saccade rate was significantly lower during AI-assisted search, further supporting a shift from broad visual scanning to concentrated processing. These findings contradict the hypothesis of banner blindness (Benway, 1998; Jerath et al., 2011). In the sense that, rather than ignoring the AI summaries as they might ignore advertisements occupying similar screen positions, participants engaged with them more deeply. This focal engagement suggests that users perceived the AI-generated content as a distinct, potentially salient information source, qualitatively different from the advertising or navigational elements that typically occupy prominent positions on search results pages. As such, the practical and design implications are significant. If users are substituting the action of scanning and clicking with the action of reading and absorbing, then potentially, AI summary functionality may serve as an information endpoint rather than merely a navigational aid. This substitution may represent the cognitive mechanism through which the traffic reductions measured in this study and documented in recent observational studies (Dewanto & Herari, 2025; Law & Guan, 2025) are produced.

The aggregate pupillometry data showed no overall difference in cognitive load between the conditions, a result that might initially suggest that AI summaries have no impact on cognitive processing. Conversely, follow-up analyses showed a more nuanced picture. During the search phase (first part of tasks), when users engaged directly with the results page, the AI condition showed a marginally significant decrease in pupil diameter, suggesting a trend toward reduced cognitive load when the summary was available. This effect disappeared during the non-search phase, when participants navigated external websites, aligning with the expectation that the summary's influence is confined to the search interface. In fact, the temporal analysis further clarified this picture by revealing a significant crossover pattern. During the early portion of the trial (approximately 0–30%), AI-assisted tasks were characterized by reduced pupil dilation relative to non-AI tasks, suggesting that the summary initially served as a cognitive scaffold that offloaded the effort of scanning and filtering multiple results (Skulmowski & Xu, 2022). In the latter portion (approximately 70–100%), this pattern reversed, with AI-assisted tasks exhibiting greater pupil dilation. This late-stage increase is consistent with a transition from passive consumption to active verification, in which users cross-reference the AI-generated summary against organic search results, engaging in integrative processing that is more cognitively demanding (Paas & van Merriënboer, 2020). This crossover pattern has potentially important theoretical implications. It suggests that AI summaries do not simply reduce or increase cognitive load; rather, they redistribute cognitive effort across the temporal arc of a search task, shifting demand from initial orientation to downstream validation. However, it should be noted that while the GAM analysis provided clear evidence for condition-specific trajectory differences, the global FDA permutation tests did not reach significance, highlighting the need for replication with larger samples.

Furthermore, the significant negative correlation between cognitive load and number of clicks appears to provide further evidence for this crossover pattern as a downstream behavioral response. This relationship suggests that when cognitive demands are higher, users click fewer external links, potentially due to investing greater processing effort in the content already visible on the results page. While this correlation was computed across conditions rather than within the AI condition specifically, it is consistent with the interpretation that the cognitive redistribution induced by AI summaries may translate into reduced clicking behavior.

The absence of significant differences in self-reported satisfaction and purchase intention, despite significant shifts in cognition, indicates a broader methodological point, that self-reported metrics alone are insufficient for evaluating the impact of generative AI on user experience (Lewandowski & Kammerer, 2021). Participants' conscious evaluations of the "lived" search experience were insensitive to changes detected by physiological measures. This disconnect may reflect the fact that the cognitive shifts observed in the current study operated below the threshold of conscious awareness, or that the habitual framing of search satisfaction prevented participants from registering a qualitative change in their experience. Regardless, the findings reinforce the view that incorporating physiological and behavioral process measures into studies of human-AI interaction is necessary, particularly when evaluating features whose effects may be gradual rather than immediate.

The present findings suggest that the transition to AI-integrated search is characterized by an immediate cognitive restructuring that precedes, and may ultimately drive, a gradual behavioral shift. While the observed decrease in clicks was not statistically significant, it is directionally consistent with the substantial reductions measured and reported in observational studies with larger samples (Law & Guan, 2025). These findings provide a plausible mechanism for this effect, such that users engage deeply with AI summaries, initially offloading orientation effort and subsequently investing in verification processes, a pattern that may progressively reduce the perceived need to visit external links as trust in AI-generated content increases.

If these results hold true across broader samples, the consequences for the digital ecosystem could be substantial. There exists a mutual dependence between search engines and content creators, in which SEs depend on the continuous production of online content while content creators depend on SE traffic for visibility and monetization. This may be disrupted if users increasingly satisfy their information needs within the search interface itself. A reduction in external traffic could diminish incentives for content creation, which would in turn degrade the training data on which AI models depend (Patil & Gudivada, 2024; Suleiman & Awajan, 2020), creating a potential negative feedback loop requiring careful attention from platform designers, policymakers, and researchers alike.

4.2 Limitations and Future Directions

Several methodological decisions in this study were made as exploratory aims, but each points toward productive directions for future research. The sample of 26 participants was appropriate for the primary aims of the study and the within-subjects design maximized sample size. A focused initial sample is also consistent with the exploratory nature of the investigation, where the goal was to identify whether cognitive and attentional signatures of AI integration were detectable in principle, rather than to estimate population-level effect sizes. Future studies should nonetheless employ larger samples to determine whether the trend reduction in link clicks reach statistical significance with greater power, and to enable more fine-grained analyses of individual differences in response to AI-generated summaries.

This study used a Figma prototype rather than a live search engine in order isolate the effect of AI summaries and minimize possible confounding variations such as personalized advertisements, dynamic search rankings, and network latency. Such sources of noise that would have complicated causal inference (Carter & Luke, 2020). This custom Search artifact also enabled precise temporal synchronization with physiological sensors, which was essential for the kind of time-resolved pupillometric analysis conducted here. That said, the prototype necessarily abstracted away the complexity of real-world information seeking, and it remains an open question whether the cognitive patterns documented here persist under more naturalistic browsing conditions. Future research integrating physiological measurement with live search environments would provide a valuable test of this generalizability, and recent advances in mobile and remote eye-tracking offer promising avenues for doing so without sacrificing temporal precision.

Furthermore, the study's focus was on capturing initial cognitive responses to a novel interface element and thus a single session was done. This was deemed important to capture at first exposure the tension between new cognitive demands and established behavioral habits. However, the theoretical framework invoked here posits a longitudinal relationship between cognitive restructuring and behavioral change, in which attentional and load patterns intensify over repeated exposure before manifesting as reductions in clicking behavior. A single session is insufficient to observe this unfolding process. Longitudinal designs tracking users over extended periods of naturalistic AI-assisted search, ideally with periodic physiological measurement, are therefore needed to test whether the early impacts on cognitive processes observed do indeed consolidate into durable shifts in search behavior, as predicted.

Finally, the task set was also constrained to two search contexts, informational and product-specific, which, while representative of common search behavior, may not capture the full range of contexts in which AI summaries operate. Also, summaries may function quite differently in high-stakes domains such as health or legal information, where accuracy scrutiny is elevated and the consequences of misplaced trust are greater. Future studies should extend the task battery to cover a wider range of search domains. Richer behavioral instrumentation, including dwell time, scroll depth, and return visit patterns, would also complement click count as an outcome measure, providing a more complete picture of how AI summaries reshape the behavioral landscape of online information seeking.

5. Conclusion

This study provides evidence that AI-generated summaries cognitively restructure search behavior before any behavioral shift becomes statistically detectable. Such behavioral changes remained stable despite the AI summaries, albeit a trend-level reduction in click-through rates, these results are consistent with the behavioral inertia predicted by Habit Theory. Still, eye-tracking data revealed a significant transition from ambient scanning to focal reading, and temporal pupillometry identified a redistribution of cognitive effort from initial orientation toward downstream verification. The significant negative correlation between cognitive load and number of clicks suggests a plausible mechanism through which this cognitive restructuring may gradually translate into reduced external web traffic, consistent with trends already documented in observational research. Critically, these findings emphasize the necessity of physiological methods for evaluating the impact of generative AI on user experience, and extended research is necessary to determine whether these early impacts on cognitive processes consolidate into durable behavioral change.

Ethics statement

This study involving humans were approved by the institutions Research Ethics Board (Projet #: anonymized). The studies were conducted in accordance with the local legislation and institutional requirements. The participants provided their written informed consent to participate in this study.

Conflict of interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The authors declare that no Gen AI was used in the creation of this manuscript.

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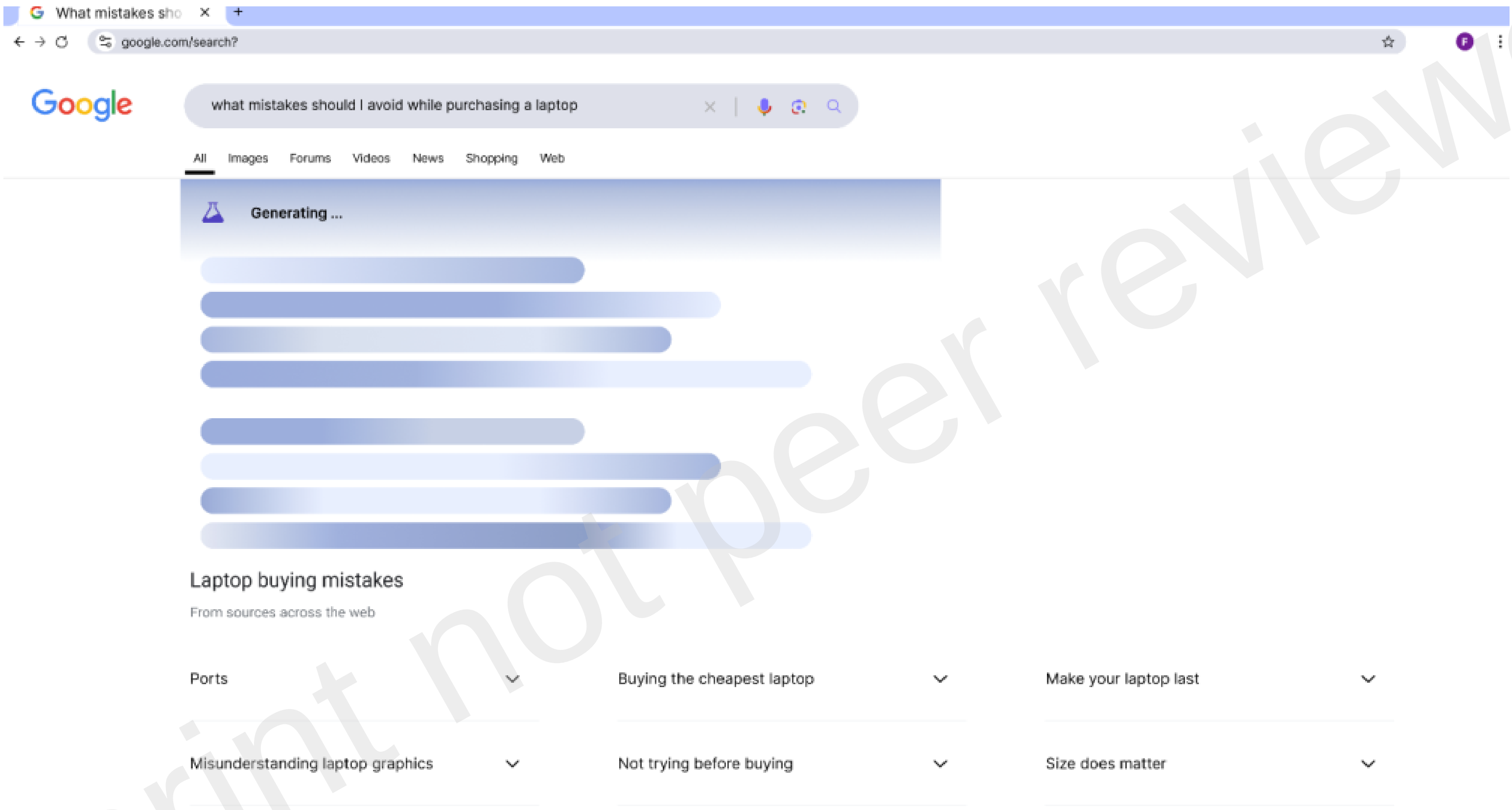

Figure 1. *Example of a summary being generated following a query in Google's AI-powered Search Generative Experience known as 'AI Overviews'.*

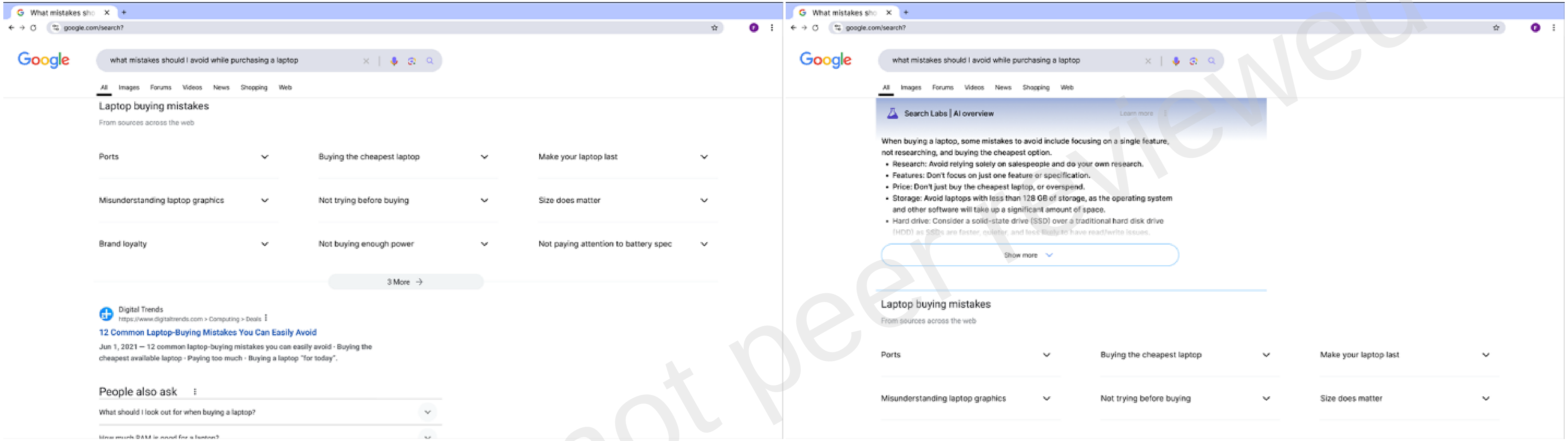

Figure 2. *left*-standard (non-AI-assisted) Google Search task for informational purposes; *right*-Google Search (AI-assisted) task for informational purposes.

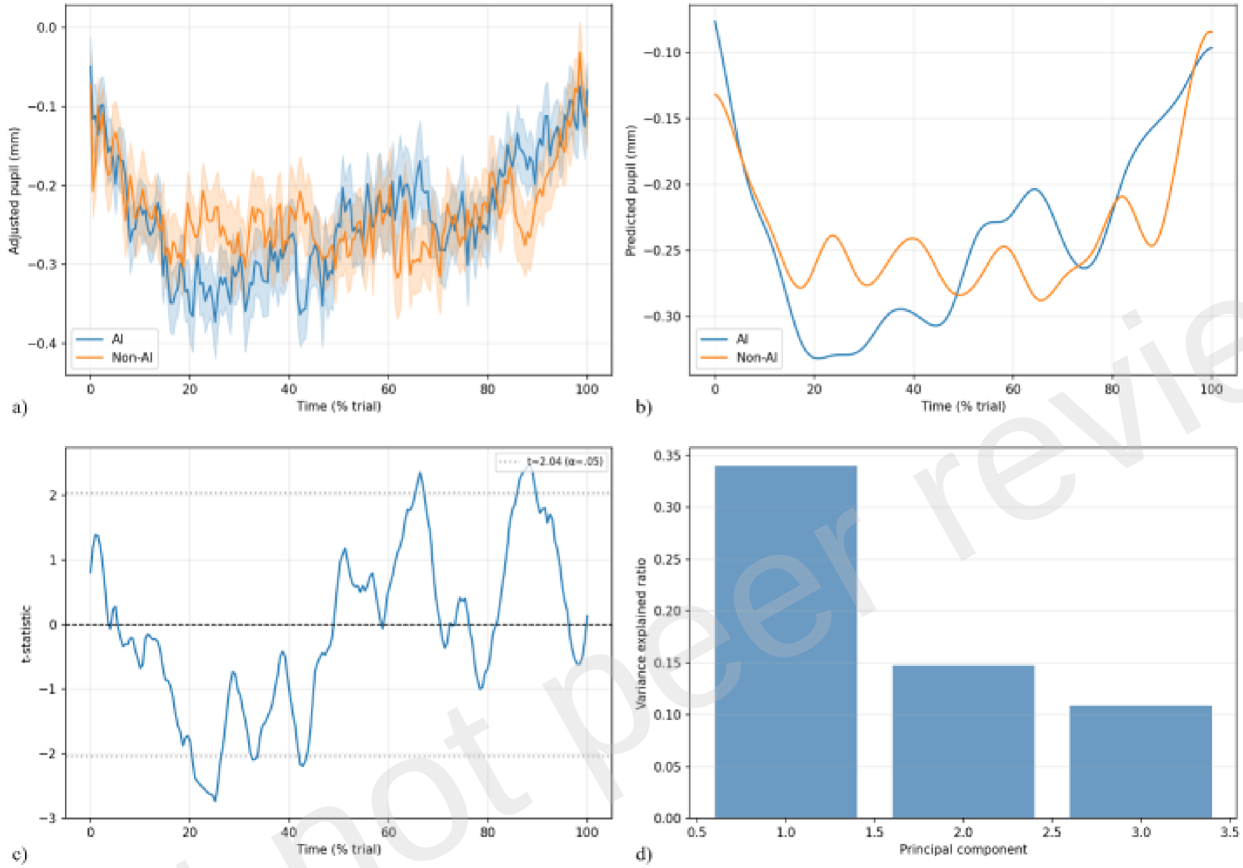

Figure 3. Time-resolved pupil dynamics across AI and Non-AI task conditions. (*a*) Mean adjusted pupil size over time (±SEM), plotted as a function of trial-normalized time. (*b*) GAM-derived pupil trajectories illustrating nonlinear time courses for each condition; the significant time × condition smooth indicates that AI and Non-AI curves differ in shape rather than in a constant offset. (*c*) FDA time-point–wise t-statistics for the AI–Non-AI difference, demonstrating early negative and late positive deviations; dotted horizontal lines depict approximate values for α = .05. (*d*) FPCA variance decomposition for adjusted pupil curves, with PC1, PC2, and PC3 corresponding to mid-trial, early, and late temporal patterns, respectively.

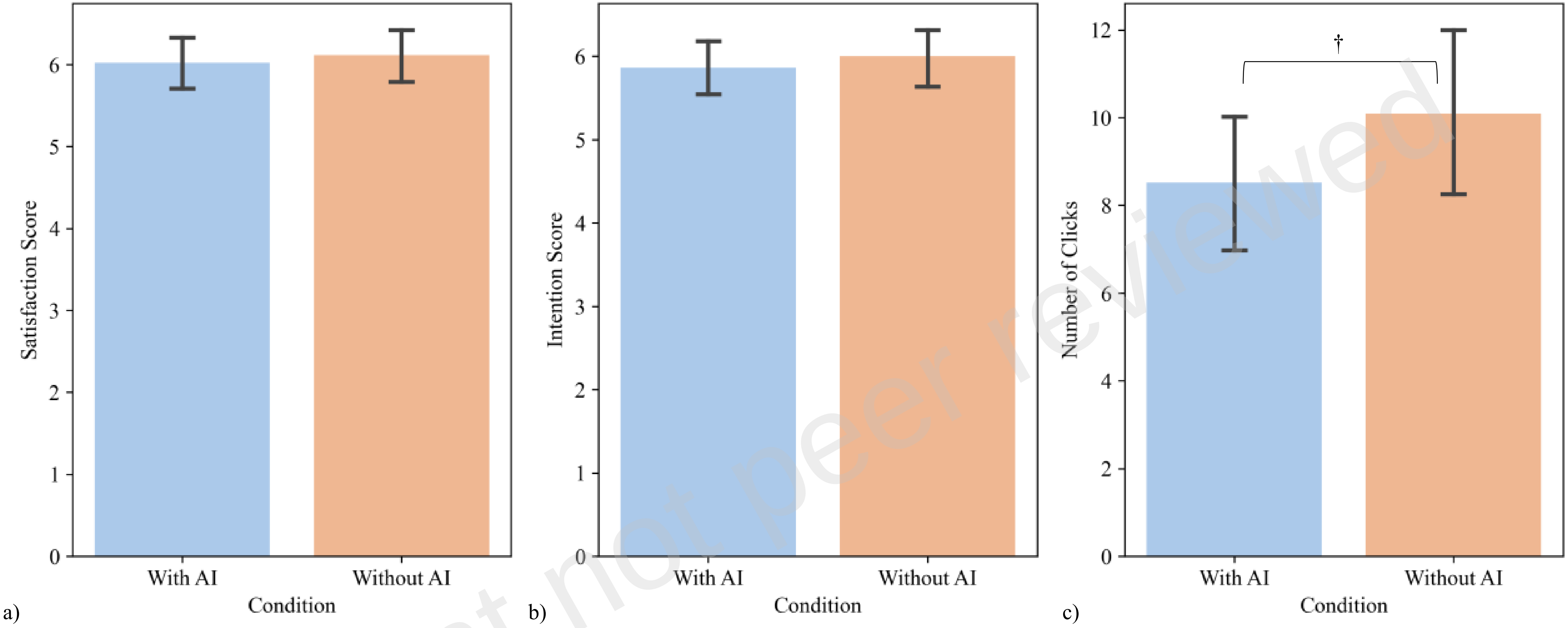

Figure 4. Average (bars) and standard deviation (errorbars) results for satisfaction (a), intention (b) and number of clicks (c) according to non-AI assisted (orange) and AI-assisted (blue) conditions, where † p < 0.10 from the Wilcoxon signed-rank tests.

Table 1. Seach tasks

Task type	Task objective	Examples
Informative	Identify important features/criteria to consider when purchasing an item	"Use Google to find what are the important features to consider when purchasing a laptop for gaming" "Use Google to find what mistakes you should avoid doing while purchasing a laptop for gaming"
Transactional	Find a Google result that leads to a site enabling purchase of items	"Use Google to buy a Macbook Air M1" "Use Google to buy a Dell XPS 15"

Table 2. Experimental conditions.

Condition	Task 1	Task 2	Task 3	Task 4
A	AI	AI	NA	NA
B	AI	NA	NA	AI
C	NA	AI	AI	NA
D	NA	NA	AI	AI

Table 3. Eye-tracking data analysis across the full task, search phase and non-search phase

Phase	Metric	AI Condition		Non-AI Condition		Test Statistics		
		M	SD	M	SD	*t*	*p*	*d*
Full Task	Adjusted Pupil Size (mm)	-0.23	0.19	-0.22	0.17	-0.2	0.842	-0.05
	Pupil range	1.67	0.41	1.57	0.51	0.87	0.389	0.2
	Pupil std	0.19	0.05	0.18	0.04	0.69	0.494	0.16
	Saccade Rate (Hz)	1.66	0.68	1.9	0.75	-1.48	0.144	-0.34
	Blink Rate (Hz)	12.08	5.69	12.72	5.79	-0.48	0.633	-0.11
Search	Adjusted Pupil Size (mm)	-0.28	0.19	-0.22	0.18	-1.85	0.078 †	-0.39
	Pupil range	1.54	0.46	1.35	0.45	1.92	0.059 †	0.43
	Pupil std	0.18	0.06	0.17	0.04	1.1	0.275	0.24
	Saccade Rate (Hz)	1.65	0.69	2.00	0.83	-2.06	.043*	-0.46
	Blink Rate (Hz)	11.92	6.23	12.69	5.49	-0.59	0.557	-0.13
Non-search	Adjusted Pupil Size (mm)	-0.22	0.2	-0.24	0.21	0.35	0.729	0.08
	Pupil range	1.5	0.46	1.49	0.55	0.05	0.961	0.01
	Pupil Std (mm)	0.19	0.06	0.19	0.05	0.45	0.657	0.1
	Saccade Rate (Hz)	1.78	0.73	1.96	0.81	-1.02	0.312	-0.23
	Blink Rate (Hz)	11.59	5.96	12.03	5.48	-0.35	0.727	-0.08

Note: $^{*}p < 0.05$, $^{\dagger}p < 0.1$.